A Controlled Comparison of Deep-Learning Paradigms for Xiangqi under a Constrained Compute Budget

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***Abstract*—Modern game-playing agents are built from three ideas validated on Go and chess at cluster scale: imitation of human experts, reinforcement learning (RL) by self-play, and tree search over a learned network. It is unknown which of these paradigms is worth its cost for *Xiangqi* (Chinese Chess) on the single consumer GPU most researchers and students actually use, whether imitation pretraining still helps the subsequent RL under that budget, and what factor ultimately bounds playing strength. Existing Xiangqi deep-RL work consists of a few single-system papers and provides no controlled, like-for-like comparison. We propose a controlled, equal-compute study that places four paradigms —classical alpha–beta search, pure-self-play Proximal Policy Optimization (PPO), imitation-initialised PPO, and AlphaZero-style Monte Carlo Tree Search (MCTS)— on a single shared Xiangqi engine, a fixed 2086-move action space, one policy–value network, and one Elo scale, so the paradigm is the only variable. We will rank the agents by fitted Elo with 95% confidence intervals, quantify the imitation→RL effect, and run a four-lever ablation (search, epochs, teacher, network size) to identify the binding constraint. A complete, tested implementation and preliminary runs over 11.6M positions from 141k human games already exist, making the study a rigorous formalisation rather than a speculative build. The expected contributions are a reusable multi-agent evaluation framework, to our knowledge the first controlled paradigm comparison for Xiangqi under constraint, and reproducibility guidance for self-play RL.**

***Index Terms*—Xiangqi, Chinese Chess, deep reinforcement learning, Monte Carlo tree search, imitation learning, proximal policy optimization, Elo rating, empirical software engineering, reproducibility.**

I. Introduction

THE dominant recipe for superhuman board-game play was established on Go by AlphaGo, which combined supervised (imitation) learning of expert moves, RL by self-play, and Monte Carlo Tree Search (MCTS) [1]. AlphaGo Zero removed the human-data stage and learned *tabula rasa* [2], and AlphaZero generalised the same algorithm to chess, shogi, and Go, replacing alpha–beta search with general MCTS [3]. These landmark results share two properties that limit their guidance for most practitioners: they were obtained on large tensor-processing clusters, and each reports a single successful pipeline rather than a controlled comparison of the paradigms it combines.

Motivation and problem. For a researcher, student, or practitioner working on a *single consumer GPU*, the practical questions are unanswered, especially for *Xiangqi* (Chinese Chess)—a culturally major but academically under-served game with a larger 10×9 board, higher branching than chess, a cannon that captures by jumping a screen, and the unusual rule that *stalemate is a loss*. The published deep-RL work on Xiangqi is recent and sparse: an AlphaZero-style MCTS player [4], a population method that tackles strategic non-transitivity [5], and a language-model approach trained with policy optimization [6]. Each builds *one* maximally strong system; none measures, under a fixed budget, how the distinct learning paradigms compare, whether imitation pretraining helps the ensuing RL, or which factor limits strength.

Significance. Deep-RL results are known to be fragile and hard to reproduce when reported from single runs without variance analysis [7]. A controlled, confidence-interval-backed comparison on one rule engine therefore yields decision-relevant, reproducible evidence that cluster-scale, single-system papers do not.

Research questions. We pose three questions, each answerable by the methodology below:

- **RQ1 (ranking):** under a fixed single-GPU budget, how do the four paradigms rank in Xiangqi playing strength on a common Elo scale?
- **RQ2 (imitation→RL):** does imitation pretraining improve subsequent PPO self-play under that budget, or can RL fine-tuning *degrade* the imitation policy?
- **RQ3 (binding constraint):** which factor—play-time search, model capacity, training data/teacher, or the learning algorithm—most limits playing strength at this scale?

Objectives and contributions. We will (i) rank four paradigms on a common Xiangqi Elo scale under matched compute; (ii) quantify the signed effect of imitation pretraining on subsequent PPO; and (iii) identify, through a single-variable ablation, whether search, model capacity, training data, or the learning algorithm binds strength. The contributions are: a reusable single-interface *multi-agent evaluation league* for heterogeneous game-playing paradigms runnable on one GPU; a controlled, equal-budget, Elo-anchored paradigm comparison for Xiangqi, including negative results; and concrete reproducibility guidance for self-play RL.

TABLE I
WHERE PRIOR WORK LEAVES THE GAP THIS STUDY FILLS.

Work	Xiangqi	Cross-paradigm comparison	Constrained GPU budget
AlphaGo family [1]–[3]	no	no	no
MCTS survey [8]	n/a	n/a	n/a
PPO [9]	no	partial	no
RL reproducibility [7]	no	no	yes
AlphaStar [10]	no	no	no
JiangJun [5]	yes	no	no
Yilmaz *et al.* [4]	yes	no	partial
Xiangqi-R1 [6]	yes	no	partial
This study	**yes**	**yes**	**yes**

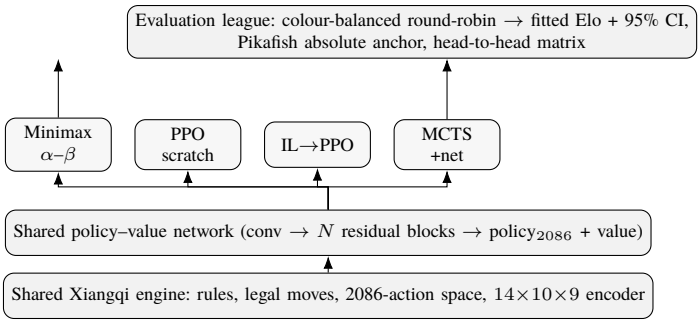


Fig. 1. The controlled apparatus. One environment, one action space, and one network are shared by all four agents; only the decision function changes, and all agents are rated on one Elo scale.

TABLE II
TRACEABILITY: EACH RESEARCH QUESTION MAPS TO AN OBJECTIVE, AN EXPERIMENT, AND A PRIMARY METRIC.

RQ	Objective	Experiment	Primary metric
RQ1	Rank the paradigms	E1 round-robin	fitted Elo + 95% CI
RQ2	IL→RL effect	E2 isolation	signed Elo delta
RQ3	Binding constraint	E3 four-lever	per-lever Elo effect

II. RELATED WORK

Deep RL with search. The AlphaGo family established the imitation→RL and search-over-network levers we compare [1]–[3]; MCTS itself is surveyed comprehensively by Browne *et al.* [8], whose PUCT family our search agent implements. Classical Xiangqi engines instead use alpha–beta search with handcrafted evaluation, the tradition our Minimax baseline represents.

Policy gradients and reproducibility. PPO [9] is our RL optimizer, chosen for stability; yet Henderson *et al.* [7] show policy-gradient results vary sharply with seeds and implementation, motivating our interval-based evaluation and a training-stability analysis.

Imitation and its ceiling. The AlphaGo pipeline initialised its policy by imitating human experts before refining it with RL [1]; a known limitation is that an imitator cannot surpass its demonstrators, the usual justification for the RL stage. Whether that stage helps *under constrained compute*, however, is untested for Xiangqi—and our preliminary data suggest it can hurt, which is precisely what a controlled study should resolve.

Populations and evaluation. AlphaStar extended self-play to a league of adapting agents rated on an Elo ladder [10]; we adopt that framing for a league of *heterogeneous* agents. For Xiangqi, JiangJun combines MCTS with Policy-Space Response Oracles to address non-transitivity, analysing >10,000 human games [5]—a reminder that single head-to-head results can mislead and that round-robins with fitted Elo are needed.

Xiangqi-specific deep RL. The directly comparable systems [4]–[6] each optimise one pipeline at either cluster scale or without cross-paradigm comparison. Table I summarises the resulting gap.

Research gap. No controlled, equal-compute, Elo-anchored comparison of classical search, pure PPO, imitation+PPO, and AlphaZero-style MCTS exists for Xiangqi under a single-GPU budget; consequently the direction of the imitation→RL effect and the binding constraint on strength are both unquantified for this game at this scale. This is primarily an *evaluation/methodological* gap, with secondary *performance* and *application* dimensions.

III. PROPOSED METHODOLOGY

Design. A controlled comparative study: the independent variable is the paradigm (or the one ablated lever); the dependent variable is Elo; everything else is fixed by a shared stack (Fig. 1), so differences in strength are attributable to the decision function alone.

Compute budget. "Equal effort" is operationalised as a matched number of self-play environment steps per RL agent on one RTX-4060-class GPU, with the budget reported beside every result so the comparison is budget-explicit and reproducible rather than open-ended.

Shared components. (1) A tested Xiangqi rule engine exposed through a Gymnasium interface, with a fixed 2086-move action space and legal masking, and a $14 \times 10 \times 9$ side-to-move-relative board encoding. (2) One AlphaZero-style policy–value network reused by imitation, PPO (actor and critic), and MCTS (priors and leaf value); its width/depth are configurable for the ablation.

The four agents. *Minimax*: alpha–beta (negamax) search with a material evaluation. *PPO-scratch*: masked PPO self-play from random initialisation. *IL→PPO*: supervised imitation of 113k human games, then masked PPO refinement from that checkpoint. *MCTS+net*: PUCT search over the network with value-head leaf evaluation, selecting moves by visit count. A Random agent is included as a floor.

Data. 141,356 legal human games (≈11.6M positions) are split at the game level into 113k/14k/14k train/validation/test sets, recorded in a committed manifest for reproducibility; an engine self-play corpus supports the distillation lever.

Procedure. Each RL paradigm is trained to a matched single-GPU budget with periodic checkpointing. *Experiment 1 (RQ1)* runs a colour-balanced round-robin among all four

agents plus Random and fits Elo with 95% confidence intervals (1) and a head-to-head matrix. *Experiment 2 (RQ2)* isolates imitation-only, IL→PPO, and PPO-scratch and reports the signed Elo deltas plus learning proxies (imitation top-1 accuracy; PPO reward and value traces). *Experiment 3 (RQ3)* varies one lever at a time—MCTS simulations $\in \{100, 200, 400, 800\}$, imitation epochs, teacher (human vs. engine distillation), and network size (1.15M vs. 4.85M parameters)—each re-entered into the round-robin, reporting each lever's Elo effect. A stability study compares fragmented, continuous, and replay-buffer-persisted training.

Evaluation metrics. The expected score of a rating difference is

$$E_A = \frac{1}{1 + 10^{(R_B - R_A)/400}}, \tag{1}$$

inverted and fitted over round-robin results to obtain each agent's rating; confidence intervals follow the delta method. Reported metrics are fitted and absolute Elo (with 95% CI), head-to-head win rates, the signed imitation→RL Elo delta, per-lever Elo effects, top-1 move accuracy, value-loss trajectories, and GPU-hour cost. Differences are called significant only when confidence intervals do not overlap; non-transitivity is handled by full round-robins and fitted Elo rather than single matches [5]. Fixed seeds, the committed manifest, and per-run config/seed/commit logging address known reproducibility pitfalls [7].

Threats to validity. Relative Elo depends on the agent pool; we mitigate this with an absolute Pikafish anchor and full round-robins rather than single matches. Few games per pair widen confidence intervals, so the decisive comparisons will be rerun with more games until their intervals separate. Findings are scoped to one game, one network family, and one GPU class, and we claim no automatic generalisation beyond that setting; the shared-stack design nonetheless keeps the comparison internally valid, since only the decision function varies between measured agents.

IV. Expected Results and Contributions

Expected outcomes. We hypothesise that (H1) AlphaZero-style MCTS attains the highest Elo; (H2) imitation pretraining raises *starting* strength but, under the constrained budget, PPO self-play does *not* raise the ceiling and may lower it; and (H3) model capacity/data—not search depth or the learning algorithm—binds strength. Preliminary runs on the existing implementation are consistent with all three and establish feasibility (Table III); these are *preliminary, confidence-interval-wide* measurements that the proposed study will formalise with more games and tighter intervals, not final results.

Contributions. (1) *Technical*: a reusable, single-interface multi-agent evaluation league for comparing heterogeneous game-playing paradigms on one rule engine, runnable on a single consumer GPU. (2) *Research*: the controlled, equal-budget, Elo-anchored comparison of classical search, PPO, imitation+PPO, and AlphaZero-style MCTS for Xiangqi; a quantified answer to whether imitation pretraining helps RL under constraint; and a multi-lever capacity-bottleneck analysis. (3) *Software engineering*: evidence that training-stability engineering—replay-buffer continuity and best-checkpoint selection—materially affects learned-model quality, yielding actionable reproducibility guidance for self-play RL. *Practical implication*: practitioners gain budget-aware guidance on which paradigm is worth its compute.

TABLE III
Preliminary feasibility measurements (relative Elo, anchor 1500). These motivate the hypotheses; the proposed study tightens and extends them.

Agent	Elo (prelim.)	95% CI
MCTS + net (AlphaZero-style)	1703	±94
Imitation-only	1557	±77
Minimax (d=2)	1541	±77
Imitation → PPO	1411	±79
Random	1287	±96

Why the outcome matters. If the hypotheses hold, the guidance is concrete: under a consumer-GPU budget, invest in search and imitation before reinforcement fine-tuning, and treat model capacity—not deeper search or more RL epochs—as the lever that raises the ceiling. A negative or mixed outcome is equally informative, because the study measures comparative strength rather than chasing a single number; either way the evidence is reproducible and budget-explicit.

Work plan. The environment, shared network, four agents, and evaluation harness are implemented and unit-tested, and preliminary runs already exist (Table III); the remaining work is (A) finalising matched-budget training of all four agents, (B) scaling Experiment 1 to tighter confidence intervals and adding the absolute Pikafish anchor, (C) completing the four-lever ablation and the training-stability study, and (D) statistical analysis and write-up. Because the measurement apparatus already exists, the residual risk is dominated by compute time rather than by feasibility.

V. Conclusion

We have proposed a controlled, Elo-anchored comparison of four deep-learning paradigms for Xiangqi under a constrained compute budget, built on a shared environment, action space, and network so that only the decision function varies. The study answers which paradigm is strongest, whether imitation pretraining helps subsequent RL, and what factor bounds strength—questions left open by a Xiangqi literature of single, mostly cluster-scale systems. Because a complete, tested implementation and preliminary results already exist, the work is a rigorous, reproducible formalisation whose contributions—a reusable comparison framework, a constrained-budget paradigm comparison for Xiangqi, and training-stability guidance—are defensible and honestly scoped, including where the evidence is negative.

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